



Subsurface structure-induced water use strategies mediate shrub-herb coexistence in soil-limited hillslopes

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Abstract

Background and aims The combined structures of soil and rock directly influence root distribution, and root zone water movement and storage. However, it remains unclear how these processes jointly affect plant water-use strategies. We aimed to examine how subsurface structure-induced root architecture governs water acquisition strategies and their implications for species coexistence.

Methods We conducted combined root traits and stable isotopes measurements on four dominant species across slope positions in a subtropical soil-limited

environment. Electrical resistivity tomography was employed to investigate the soil-rock structures at the study plots. The vertical and temporal origin (water age) of root water uptake were estimated to determine water acquisition strategies of co-existing species.

Results Slope positions had less effect on root architecture, but significantly influenced average soil thickness and water storage capacity. Diverse soil-rock combinations created multiple root niche spaces (soil layers, soil-rock interfaces, and bedrock fissures). Although plant rooting depths were generally shallow (<40 cm), these species achieve coexistence through differential occupation of these three niche spaces, enabling access to water sources of varying ages (reflected by seasonal fluctuations in water age of root water uptake, 2–63 d). Shrubs develop more extensive lateral root systems to enhance water uptake capacity and anchorage, supporting larger plant structures than herbaceous species.

Conclusion Our study reveals that in soil-limited environments, plants share water sources within the soil layer but differentially utilize water resources available at the soil-rock interfaces and within fissures. These results demonstrate that bedrock-controlled substrate heterogeneity promotes species coexistence via spatiotemporal ecohydrological niche differentiation.

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Introduction

Plant water uptake is a crucial process that connects plants to the hydrological cycle and the broader ecosystem. The way plants draw water from the soil and the interactions between plants significantly influence soil moisture, recharge of groundwater, and the overall water balance of ecosystems (Brooks et al. 2010; Sprenger et al. 2016; Zhao et al. 2024). While soils are conventionally regarded as the primary storage reservoir for root zone water and root systems, numerous studies have reported that plant can acquire bedrock-stored moisture (Liu et al. 2025; McCormick et al. 2021; Nardini et al. 2024; Schwinning 2020), which serves as critical survival-sustaining water sources for drought stress mitigation (Hahm et al. 2020; Korboulewsky et al. 2020; Nardini et al. 2021). These processes ultimately determine the distribution, abundance, and diversity of plant species within a community. Thus, understanding the plant water uptake patterns is essential for predicting how ecosystems will respond to changes in precipitation patterns, increased evaporation, and other climate-driven alterations.

Multiple factors, both above and below ground, can influence plant water uptake patterns. Among these, the underground factors primarily encompass the distribution of roots and available water, which are obviously influenced by the subsurface structures. Water uptake and transport in plants is largely dependent on root functional traits (Fu et al. 2022; Germon et al. 2020), particularly the location and function of fine roots, which determine the potential depth and extent of water uptake in plants. Numerous studies have investigated the influence of root distribution on plant water uptake patterns (Bachofen et al. 2024; Kulmatiski et al. 2020; Wang et al. 2017; Yin et al. 2024). Water uptake depth differs significantly among various plant species due to differences in root depth. Such differences can further vary with different climate regimes and temporally between seasons (Bachofen et al. 2024; Jia et al. 2017; Wang et al. 2017). In addition, spatial and temporal variations in root zone available water affect not only the amount of root water uptake, but also the root water uptake depth. Different slope positions often exhibit distinct plant available water, which significantly drives slope-dependent variations in plant transpiration and root water uptake depth (Liu et al. 2023). Meanwhile,

seasonal variations in water uptake depth have been reported for many plants in response to changes in soil water availability (Dai et al. 2023; Gessler et al. 2022; Wang et al. 2023a, b). Shifting water uptake from shallow to deeper soil layers is an important survival strategy for plants during drought periods, and at the same time, such deep-water uptake may provide some possible benefits for neighboring plants through hydraulic lift process (Belovitch et al. 2022; Dawson 1996; Montaldo et al. 2021). Thus, different species can initiate adaptive water uptake strategies for coexistence and competition in response to spatial and temporal variations in root zone water availability. For these reasons, understanding the pattern of fine root distribution and root zone water availability is crucial for plant resource acquisition and plant response to environmental stress.

The changes in root distribution and water uptake patterns mentioned above are all responses of root niches or water uptake niches to habitat adaptation, which is the key to understanding plant growth and species coexistence. Since complete root surveys are often time-consuming and destructive, in reality, stable isotope tracer approaches are more often used to investigate plant water uptake patterns and strategies (Bachofen et al. 2024; Evaristo et al. 2015; Luo et al. 2019; Wang et al. 2017; Yang et al. 2015). In particular, these approaches effectively reveal changes in vertical ecological niches (e.g., water uptake depth, or soil water vs. groundwater, etc.) and plant-plant interactions for water acquisition (Gessler et al. 2022; Kulmatiski et al. 2020; Wu et al. 2019; Zhao et al. 2024). Meanwhile, unlike the spatial origin of plant water, the temporal origin of water acquisition by plants, as another dimension of ecological niche differentiation, has received increasing attention in recent years (Benettin et al. 2025; Brinkmann et al. 2018; Evaristo et al. 2019; Luo et al. 2023; Miguez-Macho and Fan 2021; Sprenger et al. 2019). Quantifying the temporal origin in terms of the mean residence time or age of water taken up by plants can provide valuable insights into how root zone water storage and root water uptake response to changes in precipitation inputs. Nevertheless, our current understanding of how the timing of plant water uptake relates to root distribution and species coexistence is still limited.

Despite the importance of knowledge about root functional traits and soil water status for plant water adaptation strategies, most investigations on this topic

have primarily focused on the soil layers. The influence of (weathered) bedrock beneath the soil layer or soil-rock structures on root distribution has not been thoroughly investigated. A few studies have found that bedrock underlying shallow soils is restrictive for vertical extension of plant roots (Estrada-Medina et al. 2013; Wang et al. 2023a, b). However, in shallow soil overlying highly fractured bedrock, roots may extend deeper than that with solid bedrock (Estrada-Medina et al. 2013; Schwinning 2013), because these fractures provide additional space for the storage of soil and water, as well as facilitating root penetration. This phenomenon is particularly prevalent in karst environments. Rock crevices-induced differences in rooting depth result in significantly distinct distribution pattern of woody and herbaceous plant in the karst area of southwest China (Liu et al. 2019a, b). In addition, rock fragment content in soils has significant impact on root distribution, but with different response pattern between woody and non-woody species (Hu et al. 2024). Therefore, the complex structure of rock and soil primarily influences the placement and extension of plant root systems by impacting soil depth, fracture distribution, and water availability. However, knowledge gaps with respect to soil-rock-root relationship are especially large in species-rich subtropical karst ecosystem. In recent years, geophysical methods (e.g., electrical resistivity tomography, ground penetrating radar) have been effectively applied in the exploration of underground structures in karst areas (Luo et al. 2024; Tao et al. 2022), providing the possibility to characterize complex underground soil-rock structures. Combined with stable isotope technology, it is expected to uncover the complex water relationship between bedrock, soil, and plants in karst areas.

The karst area of southwest China is a typical vulnerable ecosystem, which has experienced serious rock desertification and vegetation degradation in history due to unreasonable human activities (Chong et al. 2021; Wang et al. 2019). After the implementation of vegetation restoration projects in recent decades, these areas have become significantly greener (Tong et al. 2018; Yue et al. 2024). However, field observations indicate that the vegetation composition of these greening karst areas differs significantly depending on the underlying lithology (e.g., limestone and dolomite). In the dolomite-dominated areas, the plant communities are still dominated by grasses

and shrubs after a long period of vegetation evolution. This is primarily attributable to the shallow soils and the absence of developed crevices in the subsurface, which limits the growth of deep-rooted plants, such as trees. Such plant communities with shallow root systems are likely to face greater water competition. However, in the context of climate change, the water uptake strategies of these grasses and shrubs and the mechanisms underpinning their coexistence remain unknown.

For this reason, here we investigated the variability of soil-rock structure, root traits, and isotopic compositions of precipitation and xylem water of 4 dominant plant species on the upper, middle and down slopes of a naturally restored forest, located on a typical dolomite hillslope in southwestern China. Three distinct approaches, including geophysical, isotopic and plant functional trait, were combined in this study. For the geophysical approach, electrical resistivity tomography (ERT) was used to characterize variability in soil-rock structure (e.g., soil depth, rock or fissure distribution) at the plot scale. The main objectives of this study were to assess the effect of soil-rock structures on plant root traits between different slopes, and subsequently, to analyze how these two factors jointly influence plant water uptake patterns and coexistence mechanisms. We hypothesized that soil-rock structures would affect roots distribution and their functional traits, as solid rocks in soils would occupy the soil space and impede root extension.

Materials and methods

Study site

This study was conducted at the Huanjiang Observation and Research Station for Karst Ecosystems of the Chinese Academy of Sciences, located in a typical karst watershed (1.14 km²) in Guangxi province, southwestern China (24°43′58.9″–24°44′48.8″N, 108°18′56.9″–108°19′58.4″E). This study area has a subtropical humid monsoon climate with a mean annual precipitation of 1390 mm and mean annual temperature of 18.5 °C. This area exhibits significant seasonal variations in precipitation, with distinct wet (April to September) and dry (September to next March) seasons. The bedrock type is mainly dolomite.

Soils, developed from dolomite, are relatively shallow and generally patchily distributed, which results from the presence of numerous rock outcrops on the surface, high proportion of rock fragments in soils, and shallowly buried bedrock within the subsurface. Vegetations on the hillslopes are dominated by shrubs and tussocks although this plant community has been recovered from disturbances about 40 years. Trees are scattered in deep soil habitats on the slope or downhill.

The experimental sites were located on a south-facing hillslope in this small watershed, with a mean slope of 34.2°. A total of six sample plots (20x20 m²) were established at the upper, middle and down slope positions along two transects. The two transects were selected in typical shrub-grassland mosaics on this south-facing slope, separated by 200 m. Based on field vegetation survey, four dominant plant species in each plot were selected, including two woody plants (*Vitex negundo* and *Tirpitzia ovoides*), one herbaceous plant (*Neyraudia reynaudiana*), and one fern species (*Dicranopteris linearis*). Further details regarding the study species can be found in Table 1. Note that only *Vitex negundo* and *Dicranopteris linearis* are available for investigation in the downslope plot.

Water isotope sample collection and stable isotope analysis

Plant xylem samples were collected monthly between April and November 2021, with five

replications for each species at the sample plots. For woody plant, we cut twigs on sunny branches. For herbaceous and fern species, we collected stems (or culms) from the base of the plant as such species usually grow in clusters with many individual main culms. Barks of these twig/stem were immediately removed before being stored into glass vials, then sealed with parafilm and brought back to the laboratory for further analysis. Water samples of precipitation were collected after each precipitation event using a standard precipitation collector. Precipitation amount as well as other climate variables (temperature, relative humidity, wind speed, etc.) was measured at a 30-min interval using an automatic weather station installed at the study site.

Plant xylem water was extracted using a water extraction system (LI-2100, LICA United Technology Limited, China) based on the cryogenic vacuum distillation method. Isotopic compositions of xylem water and precipitation ($\delta^{18}\text{O}$ and $\delta^2\text{H}$) were analyzed with a liquid water isotope laser spectroscopy instrument (Model DLT-100, LGR Inc.) at the Key Laboratory of Agro-Ecological Processes in the Institute of Subtropical Agriculture, the Chinese Academy of Sciences. $\delta^{18}\text{O}$ and $\delta^2\text{H}$ values are expressed in delta notation (δ) as

$$\delta^{18}\text{O}(\text{or } \delta^2\text{H}) = (R_{\text{sample}}/R_{\text{standard}} - 1) \times 1000 \quad (1)$$

where R_{sample} and R_{standard} are the $^{18}\text{O}/^{16}\text{O}$ and $^2\text{H}/^1\text{H}$ ratios of the sample and Vienna Standard Mean

Table 1 Basic information about the selected species in this study

Species	Leaf phenology	Slope position	BD (cm)	H (cm)	MRD (cm)
<i>Vitex negundo</i>	Deciduous shrub	Upslope	2.89 ± 1.35 ^a	252.5 ± 58.9 ^a	30.3 ± 2.9 ^a
		Midslope	2.62 ± 1.72 ^a	255.9 ± 68.8 ^a	31.5 ± 4.8 ^a
		Downslope	3.91 ± 1.24 ^b	342.8 ± 82.9 ^b	39.6 ± 4.2 ^a
<i>Dicranopteris linearis</i>	Evergreen fern	Upslope	0.81 ± 0.14 ^a	131.0 ± 19.4 ^a	7.9 ± 1.7 ^a
		Midslope	0.83 ± 0.11 ^a	157.2 ± 30.7 ^a	7.1 ± 1.4 ^a
		Downslope	0.84 ± 0.15 ^a	178.0 ± 32.9 ^b	9.7 ± 1.4 ^a
<i>Tirpitzia ovoides</i>	Evergreen shrub	Upslope	2.63 ± 1.26 ^a	251.1 ± 61.8 ^a	28.1 ± 1.9 ^a
		Midslope	2.41 ± 0.81 ^a	246.9 ± 50.0 ^a	31.7 ± 2.1 ^a
<i>Neyraudia reynaudiana</i>	Evergreen herb	Upslope	0.97 ± 0.16 ^a	309.5 ± 9.3 ^a	28.0 ± 2.9 ^a
		Midslope	1.10 ± 0.17 ^a	291.3 ± 48.3 ^a	29.5 ± 3.9 ^a

BD is the average basal diameter, H is the average height, and MRD is the mean maximum rooting depth. The values are data ± SD, and the different superscript letters indicate significant differences between slope positions at the $P < 0.05$

Ocean Water (V-SMOW) standard, respectively. The accuracy of the instrument was $\pm 0.6\text{‰}$ for $\delta^2\text{H}$ and $\pm 0.2\text{‰}$ for $\delta^{18}\text{O}$. Given the potential influence of organic contaminations (e.g., methanol and ethanol) present in extracted xylem water samples on isotopic signature analysis, requisite corrections were implemented following the method we used before (Luo et al. 2019).

Determining the vertical and temporal origin of root water uptake

For vertical origin of root water uptake, we simply inferred from the depth of root distribution rather than estimating water uptake depth based on isotopes of water sources from different depths, because it is not easy to obtain representative water sources at different depths in such complex soil-rock structure habitat. This was not done because, on the one hand, these plant roots are not only distributed in shallow soils, but also penetrate fissures for water uptake. On the other hand, frequent soil sampling in such soil-limited habitats may result in significant disruption to the root zone. Thus, the maximum depth of root water uptake was estimated based on the observed maximum fine root distribution depth in this study.

For temporal origin of root water uptake, water age of root water uptake was calculated based on a piecewise linear mixing water age model we proposed before (Luo et al. 2023). It is well known that isotopic compositions of plant xylem water generally reflect the integrated isotopic compositions of water in the root zone where roots acquire water for transpiration. That is to say, age of transpired water (xylem water) can represent the mean age of water taken up by roots. We assumed that precipitation is the main source of water replenished in the effective root zone where roots acquire water for transpiration, and that the effect of water storage within plant xylem on isotopic compositions of root water uptake is negligible. Thus, mean water age of root water uptake (A_r) was calculated as follows based on isotopic compositions of precipitation and xylem water:

$$A_r(t_i) = A_p(t_i) \times \beta + [A_r(t_{i-1}) + \Delta t] \times (1 - \beta) \quad (2)$$

$$\beta = (\delta_{x,i} - \delta_{x,i-1}) / (\delta_{\Sigma P} - \delta_{x,i-1}) \times 100 \quad (3)$$

$$A_p(t_i) = t_i - \sum_1^k (t_k \times P_k) / \sum_1^k P_k \quad (4)$$

where $A_r(t_i)$ and $A_r(t_{i-1})$ are mean water age of root water uptake at time steps t_i and t_{i-1} , $A_p(t_i)$ is the precipitation-weighted residence time of newly replenished water to root zone at time step t_i , quantifying the duration between meteoric water entry into the root zone and its subsequent root uptake. Here β is the contribution proportion of precipitation to root water uptake, which is to determine whether and how much new water (precipitation) enters the root zone water acquired for transpiration between two consecutive sampling events. $\delta_{x,i}$ and $\delta_{x,i-1}$ are isotopic compositions of plant xylem water at time steps t_i and t_{i-1} , $\delta_{\Sigma P}$ represents the volume-weighted mean value of isotope values of precipitation between two consecutive sampling events. The k is the number of precipitation events, t_k and P_k are the occurrence time and amount of k^{th} precipitation event, respectively. Here, $\Delta t = t_i - t_{i-1}$. In this study, only $\delta^{18}\text{O}$ values were employed for A_r calculation, given that no significant fractionation of oxygen isotopes during root water uptake has been reported. Meanwhile, the model includes a correction for evaporative enrichment. More details on this water age model can be found in Luo et al. (2023). In addition, and more importantly, we find that the evaporative enrichment in soil water at the study site was minimal based on our previous isotopic measurements in a similar habitat at the same site, because the isotopic values of soil water were distributed along the precipitation, and their fitted slopes were very close (Fig. S2).

Measurement of fine root traits

Root samples were collected in the study plots on the three slope positions. For each individual plant (five replicates), the root systems were carefully removed from the soil, rock fragments and fissures using a geological hammer and soft-bristled brush. During root excavation, we started from the main root and continuously traced and excavated along the root extension direction. When roots penetrated into the

soil-rock interface or bedrock fissures, we carefully used tools to break open the rock in order to extract the intact root samples from the fissures. Then, the intact fine root (diameter ≤ 2 mm) strands were sampled by tracing coarse roots from the stem, placed in ziplock bags, and transported to the laboratory at the nearby (about 500 m) research station for further analysis. Meanwhile, the maximum vertical rooting depth and maximum horizontal extension distance of lateral roots were measured during root excavation. The root systems of all the study species exhibit clear and distinctive characteristics from those of their surrounding plants in terms of color and shape, facilitating their identification during root excavation. The width of root excavations was determined by the distribution characteristics of the plant root systems, and typically ranged from approximately 1 to 3 m in width.

Prior to measuring root traits, it is essential to undertake a meticulous cleaning of the root system. Then, these roots were scanned with an Epson Perfection V850 scanner at 600dpi. After scanning, these fine roots were oven dried at 60 °C for at least 48 h and subsequently weighted. A common used software, *R_{HIZO}V_{ISION} E_{XPLORER}*, was used to measure root traits based on the obtained root images, which can provide more than 10 traits for better understanding root function. In this study, we focused only on five root traits related to water acquisition, such as root depth, root branching frequency (the number of root tips divided by the total fine root length, mm^{-1}), specific root length (fine root length divided by root dry weight, m g^{-1}), specific root area (root surface area divided by root dry weight $\text{cm}^2 \text{g}^{-1}$) and root tissue density (root dry weight divided by root volume, g cm^{-3}). Meanwhile, when analyzing the root traits with the *R_{HIZO}V_{ISION} E_{XPLORER}*, the root diameter was set in the following four levels: 0–0.5 mm, 0.5–1.5 mm, 1–2 mm, and > 2 mm. Then, the total root length for each root diameter range will be provided. To reflect the distribution characteristics of absorptive roots, the following indexes were calculated:

$$R_{\text{root} < 1 \text{ mm}} = L_1 / L_2 \times 100 \quad (5)$$

where $R_{\text{root} < 1 \text{ mm}}$ (%) is the ratio of root length with diameter ≤ 1 mm to the total fine root length, L_1 and L_2 are the total length of roots ≤ 1 mm and ≤ 2 mm

in diameter, respectively. Although the definitions of fine roots and absorbing roots are different, here $R_{\text{root} < 1 \text{ mm}}$ can reflect most of the absorptive roots for the study species.

Furthermore, based on root distribution patterns in karst environments, we classified the root niche space into three compartments: the soil layers, soil-rock interfaces, and bedrock fissures. During root excavation, we documented root distribution across these three compartments and roughly estimated the proportion of roots allocated to each niche separation zone based on measured root length.

ERT survey

To investigate the spatial variability of the soils and bedrock, a widely used resistivity-meter, Syscal Pro unit (IRIS Instruments, France), was employed for the ERT survey in this study. This acquisition system has 10 channels and 72 electrodes. Three ERT transects were set up for each plot, and the distance between the adjacent transects was about 5 m. The Wenner array was chosen for resistivity measurements because it is relatively sensitive to vertical changes in the subsurface resistivity. All electrodes were forced 10–15 cm below the surface with an inter-electrode spacing of 0.5 m. Prior to each set of measurements, a resistivity check process was conducted to ascertain the correct connection and optimal contact between the electrodes and the ground (resistance < 2 kOhm). In the event of inadequate contact, a small quantity of saltwater was added around the electrodes to enhance the contact resistance. The ERT survey was conducted on June 2023. Meanwhile, the locations of electrodes as well as outcrops were recorded during the resistivity data collection.

The Prosys II software was used to analyze the measured apparent resistivity data. Outliers and negative apparent resistivity values were removed. Then, data inversion was conducted using the software ViewLab3D, which is a useful tool for geophysical data inversion and visualization. The inverted 2D resistivity images of the observed transects were produced, with a maximum number of inversion iterations of 7 and a tolerance (stop criterion) for inversion iterations of 0.001.

Additionally, the repeatability uncertainty in the measurements was evaluated by conducting three repeated measurements at the same location in a day.

Due to the significant differences in water content, porosity and structure between soils and rocks, subsurface electrical resistivity generally exhibits substantial variations between soils and rocks. Thus, the distribution of soils, large rock fragments and soil–bedrock interfaces can be detected based on such a high resistivity contrast. Here, soil–rock interfaces were systematically identified according the approach in Luo et al. (2024). In order to gain further insight into soil–rock structure, a series of metrics were quantified, including the mean soil thickness (m), equivalent soil thickness (the thickness of the total soil that has evenly laid in the plot, m), soil stock (the volume of soil in the given three-dimensional space, $\text{m}^3 \text{m}^{-3}$), and soil–rock ratio (the ratio of estimated soil volume to rock volume within the sample plot, %) within the study plot (the observed vertical depth is about 4 m). Meanwhile, a hand auger with a diameter of 5 cm and a Shaw Backpack Geological Exploration Rig were used to collect soil columns and bedrock core samples. This helps to obtain the information of soil depth, the soil–rock interface and bedrock weathering intensity, providing empirical validation for ERT-inferred results of soil–rock structures. Soil samples collected by the hand auger were used for measurements related to soil moisture.

Meanwhile, plant available water is essential for root water uptake and plant coexistence. In order to compare the differences in subsurface water supply capacity of different slopes, plant available water capacity was calculated based on the difference between soil water content at field capacity and wilting point multiplied by mean soil thickness, reflecting the amount of stored soil water available for plant use. In this study, we focused only on plant available water capacity in soils because (1) most of the roots of these species on different slope positions are relatively shallow and clustered in soil layer, (2) there are reliable observations of soil thickness, and (3) the roots' random interspersions through fissures poses a challenge for measuring the moisture within these fissures. The values of field capacity

and wilting point were measured from soil samples in the three plots. The wilting point was measured using a pressure plate extractor at soil matric potential of -1.5 MPa , and the value was about $0.18 \text{ cm}^3 \text{cm}^{-3}$.

Statistical analysis

Root trait variations across slope positions (upper, middle, and down slopes) and between species were evaluated using one-way ANOVA with Tukey's post-hoc test. Normality and homogeneity of variances were verified via Shapiro–Wilk and Levene's tests, respectively. Linear regression models quantified the influence of root functional traits (specific root length, branching frequency, specific root area, root tissue density, root depth) and plant available water capacity on average water age for the four species. The significance threshold was uniformly set at $P < 0.05$, with 95% confidence intervals calculated via bootstrapping. All analyses were conducted in Origin2024 software (OriginLab, USA).

Results

Variations in soil–rock structures, root system architecture and plant size at different slope positions

The soil layer in the study plots was relatively shallow and thin (Table 2 and Fig. 1). Not surprisingly, these indexes related to soil thickness or soil stock exhibited a significant ($P < 0.05$) increasing trend from upslope to downslope. Meanwhile, the continuity of the soil layer was frequently disrupted by the outcrops or relatively large rocks in soils. There were no deep and large fissures (with a width at the decimeter scale) both on the surface and subsurface, but the degree of weathering of the bedrock was very heterogeneous (Fig. 1). Areas of highly-weathered bedrock were very densely fractured, even showing layers of powdered sand mixed with large amounts of small gravel. Spatially, the highly weathered bedrock areas were distributed in patches or locally in continuous layers (Fig. 1a and b). Such areas provide space for water storage and root extension. Weakly-weathered bedrock dominated the structure of the profile, which had the largest distribution area, and

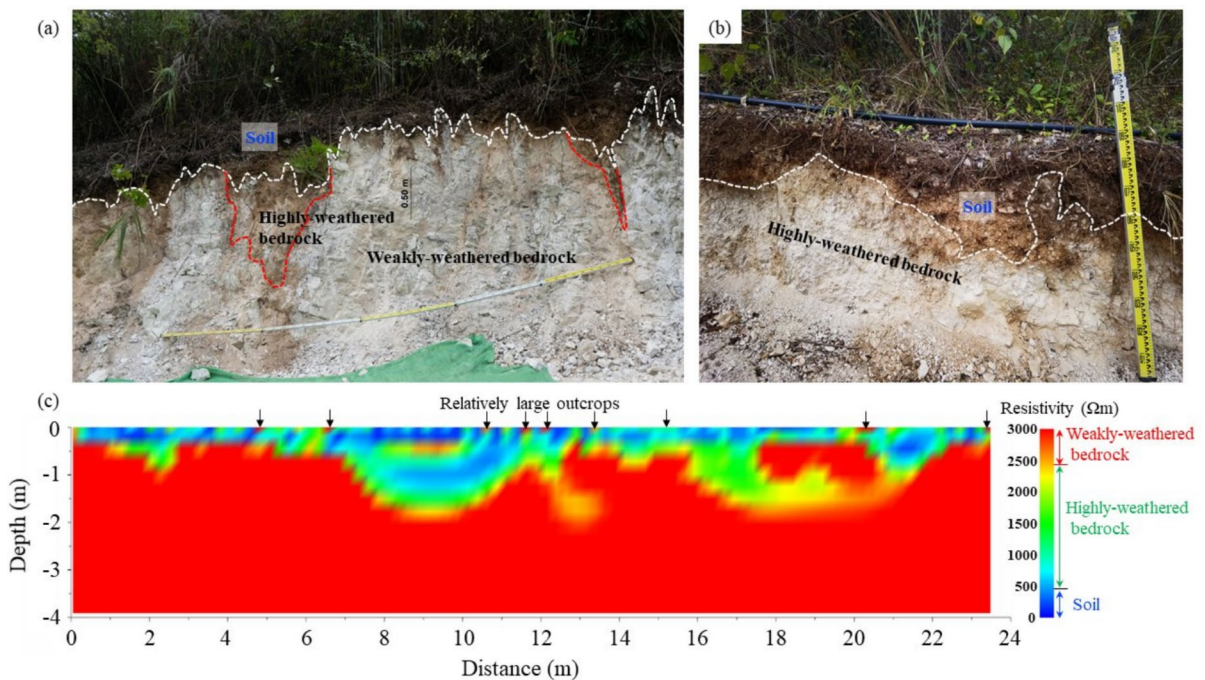
Table 2 Comparison of soil-rock structure indexes and plant available water capacity of different slope positions

Index	Downslope	Midslope	Upslope
Mean soil thickness (m)	0.50 ± 0.07	0.24 ± 0.05	0.12 ± 0.02
Equivalent soil thickness (m)	0.41 ± 0.02	0.14 ± 0.04	0.10 ± 0.01
Soil stock ($\text{m}^3 \text{m}^{-3}$)	0.101 ± 0.004	0.034 ± 0.010	0.026 ± 0.002
Soil-rock ratio (%)	11.28 ± 0.47	3.57 ± 1.04	2.66 ± 0.19
Plant available water capacity (mm)	64.8 ± 8.8	27.4 ± 5.1	13.5 ± 2.5

the rock volume is large and relatively intact, with locally small cracks. In addition, the plant available water capacity, estimated based on soil layer, differed significantly ($P < 0.05$) among slope positions, with 13.5, 27.4 and 64.8 mm on the upslope, midslope and downslope respectively (Table 2).

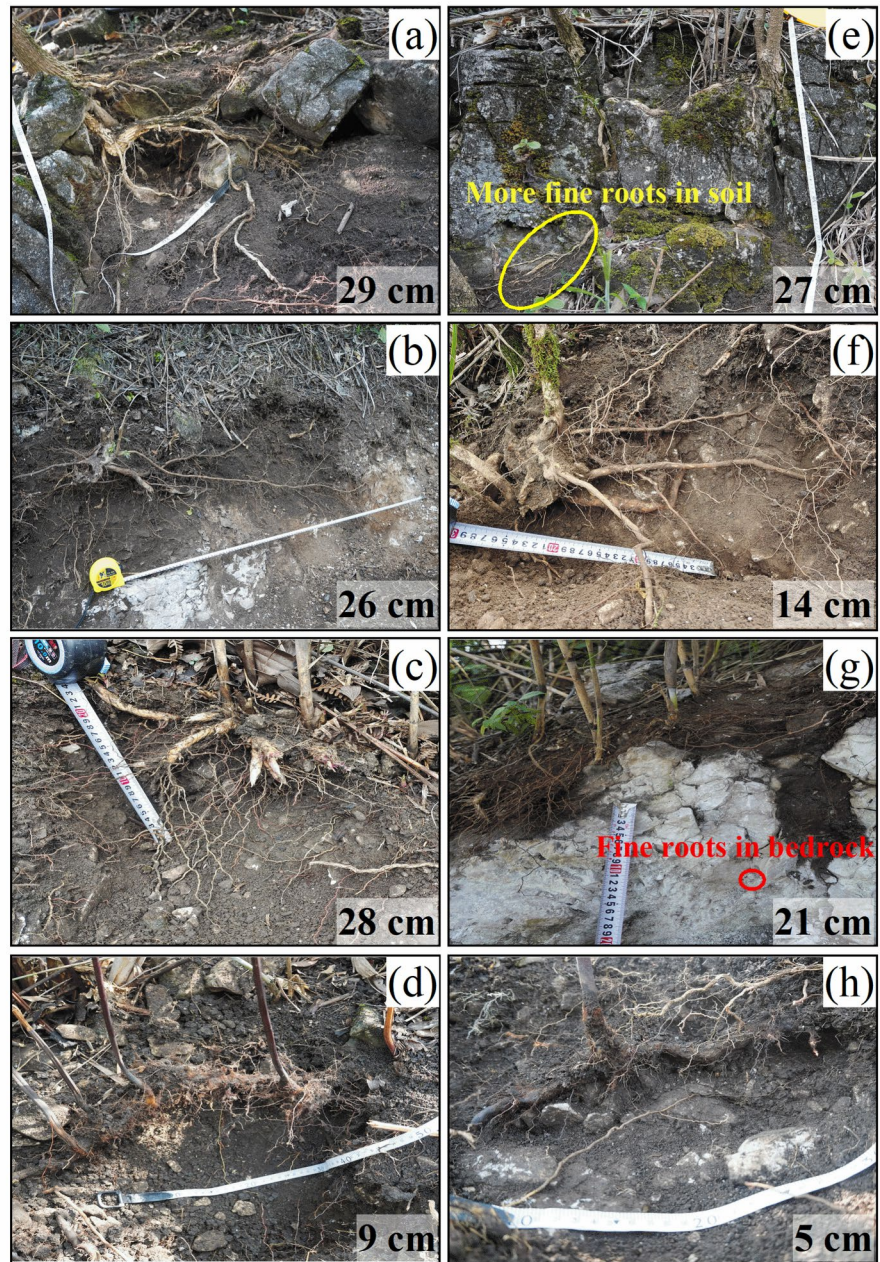
Due to the limited soil layer and insufficient development of deep vertical fractures, most of the roots in the plant community are concentrated within the upper 0–20 cm layer, with only a small fraction extending into fractures within the highly weathered bedrock (Fig. 1 and Fig. S1). The four investigated species exhibited relatively shallow root distributions, but their root morphology varied markedly (Fig. 2).

Despite differences in soil-rock structures, the root systems of *T. ovoidea* and *V. negundo* were predominantly lateral, with a horizontal spread reaching up to 100–230 cm, exceeding 3–5 times the canopy radius. Such a pattern of root distribution helps to increase the water absorption area and enhance anchorage for plants in shallow soil habitats, which in turn supports larger above-ground plant size (reflected by $\text{BD} \times \text{H}$, see Table 1). The root systems of the above two species were usually scattered in soils, rock-soil interfaces or fissures, particularly developing a greater number of fine roots when encountering soil (Fig. 2a, b, e and f). The frequency of coarse root curvatures in *T. ovoidea* was higher than that in *V. negundo*, as *T.*

**Fig. 1** Examples of two exposed soil-rock profiles (a and b, approximately 10 m from the study plots) and a 2D resistivity profile measured in the plot located on the midslope (c).

The white dashed line in (a) and (b) indicates the soil-rock interface, and the red dashed lines show the interface between highly-weathered bedrock and weakly-weathered bedrock

Fig. 2 Root distribution patterns of *T. ovoidea* (a, e), *V. negundo* (b, f), *N. reynaudiana* (c, g) and *D. linearis* (d, h) in contrasting soil depth habitats: > 20 cm (left column) vs. < 10 cm (right column), showing differences in root architecture caused by soil-rock structures. The values on these photos indicate the observed maximum rooting depth



ovoidea was more adapted to the rocky soil habitat. The observed maximum root depths for *T. ovoidea* and *V. negundo* were 34 and 45 cm, respectively, with slight variations across different slope positions.

In contrast, *N. reynaudiana* and *D. linearis* possessed short and long rhizomes that extend laterally, respectively, but the horizontal spread of the roots was relatively limited (5–30 cm). Such root systems result in above-ground plants that are usually

distributed in clumps or clusters, and are relatively small in size. Notably, the roots of *N. reynaudiana* were relatively thick and rigid, predominantly penetrating vertically into the soil in habitats with a soil depth > 20 cm (Fig. 2c), or aggregating and extending along the bedrock surface under extremely shallow soil conditions (Fig. 2g), with minimal fine roots intrusion into fractures. However, due to the shallow bedrock depth in this region (shallow soil

layer), the observed maximum root depth of *N. reynaudiana* was approximately 34.6 cm. Meanwhile, *D. linearis* had a predominantly lateral root distribution, with fine roots covered in rust-colored trichomes (root hairs), and the root depth was the shallowest among the study species (with the observed deepest root reaching only about 11.3 cm on the downslope). In short, soil-rock structures significantly influence root architecture by modulating soil thickness, the distribution of soil-rock interfaces, and fractures, thereby governing root extension and colonization in both vertical and horizontal dimensions. Furthermore, the $R_{\text{root}<1\text{ mm}}$ varied among these species, with mean values of 80.0, 85.5, 72.5, and 98.6% for *T. ovoidea*, *V. negundo*, *N.*

reynaudiana and *D. linearis*, respectively, reflecting different ability of the roots to acquire soil resources.

Root traits and root niche partitioning

Figure 3 illustrates the variability in fine root traits that are associated with the acquisition of water and nutrients across different species and slope positions for the four species under investigation. For the same species on different slopes, significant differences ($P < 0.05$) were observed only in specific root area and root tissue density of *V. negundo* between the upslope and downslope, the specific root length of *D. linearis* between the midslope and downslope, and

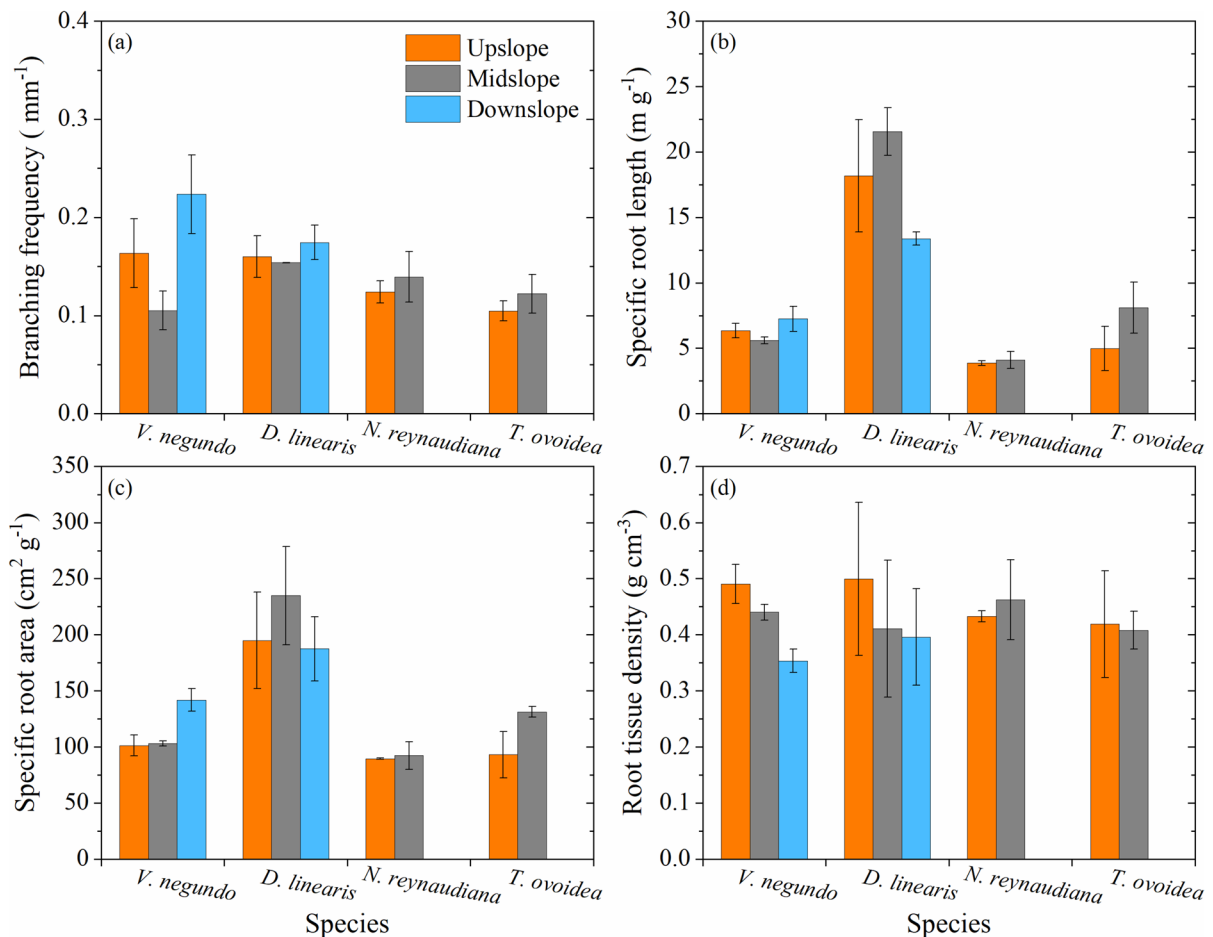


Fig. 3 Variations in branching frequency, specific root length, specific root area and root tissue density of the four species on different slopes

the specific root area of *T. ovoidea* between the mid-slope and upslope; no other root traits demonstrated significant discrepancies across slopes ($P > 0.05$). For interspecific comparison, the branching frequency, specific root length and specific root area of *D. linearis* were significantly ($P < 0.05$) distinct from those of *T. ovoidea* and *N. reynaudiana*. There were no significant differences ($P > 0.05$) of the four root traits among *V. negundo*, *T. ovoidea*, and *N. reynaudiana*.

Such similarities in fine root functional traits among these species are likely increase competition for resources (e.g., water and nutrients). However, plant coexistence is maintained through spatial niche partitioning in root distribution (Fig. 2 and Fig. 4). Plants at different slopes tend to direct their most roots towards resources in the soil as roots in soil layers account for 57–99% for different species. Although root systems of *D. linearis* concentrated in the top 10 cm soil layers (>97%), the other three co-occurring species (*V. negundo*, *T. ovoidea*, and *N. reynaudiana*) exhibited root colonization in both the soil-rock interfaces and bedrock fissures. For

N. reynaudiana, on average, 14.9% of the root system was distributed at the soil-rock interfaces, with 3.8% in bedrock fractures. In contrast, *V. negundo* and *T. ovoidea* exhibited a mean value of 21.5% and 36.5% of roots at the soil-rock interfaces, respectively, while 3.3% and 5.2% of their roots were found in bedrock fractures (Fig. 4). Importantly, *V. negundo* and *T. ovoidea* developed extensive lateral root systems (Fig. 2), thereby achieving clear niche separation and maximizing resource utilization across all root niche compartments.

Spatial and temporal origin of root water uptake

The maximum depth of root water uptake for the four species was relatively shallow based on the observed maximum rooting depth. For *V. negundo*, the maximum depth of root water uptake varied between 30 and 40 cm from the upslope to the downslope. For *D. linearis*, root water uptake was limited to the top 10 cm, with no significant ($P > 0.05$) differences between slopes (see maximum rooting depth in the

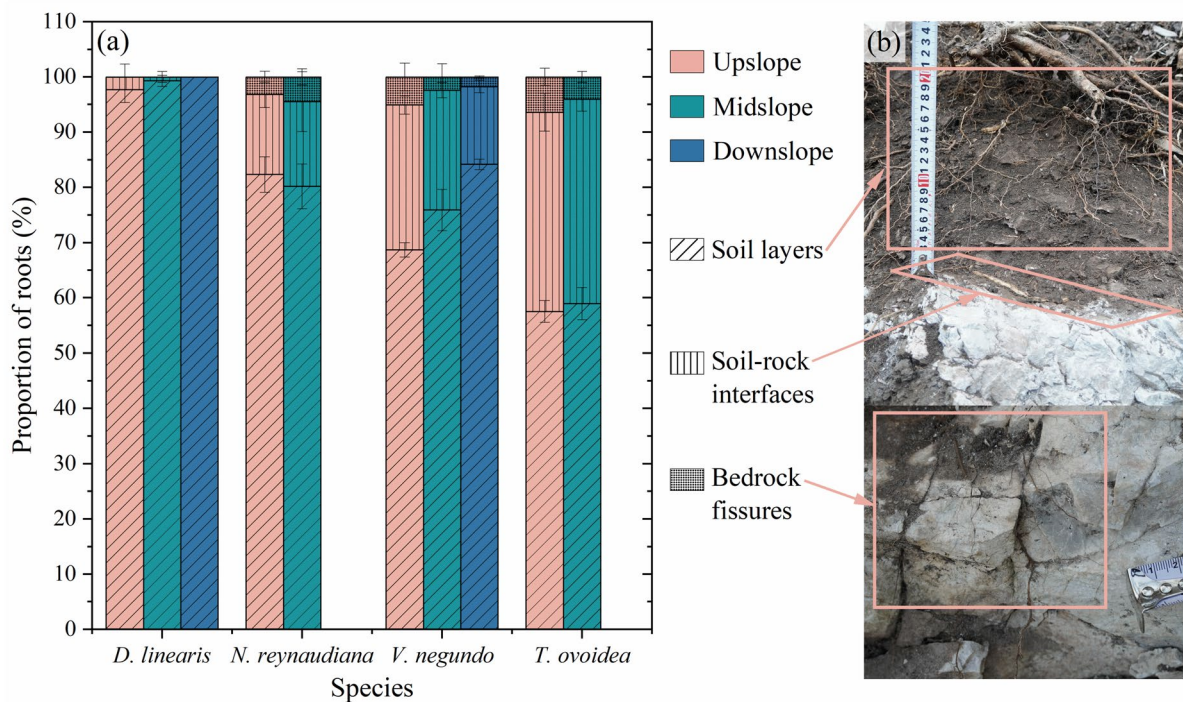


Fig. 4 Percentage distribution of root systems across three ecological niches: soil layers, soil-rock interfaces, and bedrock fissures. The proportion of roots (a) were estimated based on

root length observed in the three root niche compartments (b), and these niche spaces can vary both vertically and horizontally

Table 1). The maximum depth of root water uptake for *T. ovoidea* and *N. reynaudiana* was relatively similar, with 32 cm (28 cm) and 30 cm (28 cm) on the midslope (upslope) for *T. ovoidea* and *N. reynaudiana*, respectively (Table 1).

Figure 5 shows a comparison of water age of root water uptake between species on different slopes. Water age of root water uptake for these species increased gradually during the early part of the growing season (from April to June, except for *V. negundo* and *T. ovoidea* on the upslope), indicating that plant root zone water was gradually replenished by precipitation in the previous month or two. Then, water age declined from June to September due to relatively high root zone water replenishment by increased precipitation. Subsequently, water age increased as new water input (precipitation) decreased. Temporally, plants exhibited species-specific variations in the age of water they extracted via roots (Fig. 5a). In addition, species on the downslope generally exhibited higher values of water age. Although there were species-specific differences in how they respond to seasonal changes in water uptake, the overall averaged water age of root water uptake was relatively consistent across species. The averaged water age did not differ significantly ($P > 0.05$) between species and between slopes (Fig. 5b), ranging from 22.3 to 33.9 d, with slightly higher values on the downslope. This indicates that despite potential niche differences, all studied plants are ultimately utilizing a common, frequently replenished water pool shaped by the karst environment.

Of the factors we examined, increased root branching frequency significantly ($P = 0.048$) correlated with averaged water age of root water uptake (Fig. 6a), which explains 41% of the variation in averaged water age. This suggests that a higher branching frequency helps fine roots to access water in different niche spaces, which may consist of different water ages. Root tissue density was significantly ($P = 0.049$) negatively correlated with averaged water age in the four species, indicating that plants with higher root tissue density on midslope and upslope have a higher ability to penetrate rock fissures, and thus predominantly access rapidly recharged water reservoirs (younger water age) linked to frequent precipitation in this humid area. The other three root traits exerted a weak influence on averaged water age of root water uptake ($P > 0.05$, Fig. 6). Moreover, averaged water age was significantly ($P < 0.001$) influenced by plant available water capacity (Fig. 6f), indicating that soil thickness, controlled by soil-rock structure, has a strong impact on plant available water resources in space and time.

Discussion

Soil-rock structure impacts the pattern of root system architecture

Despite the differences in lateral root morphology of the four investigated species, the depth of root distribution was relatively shallow on different slope

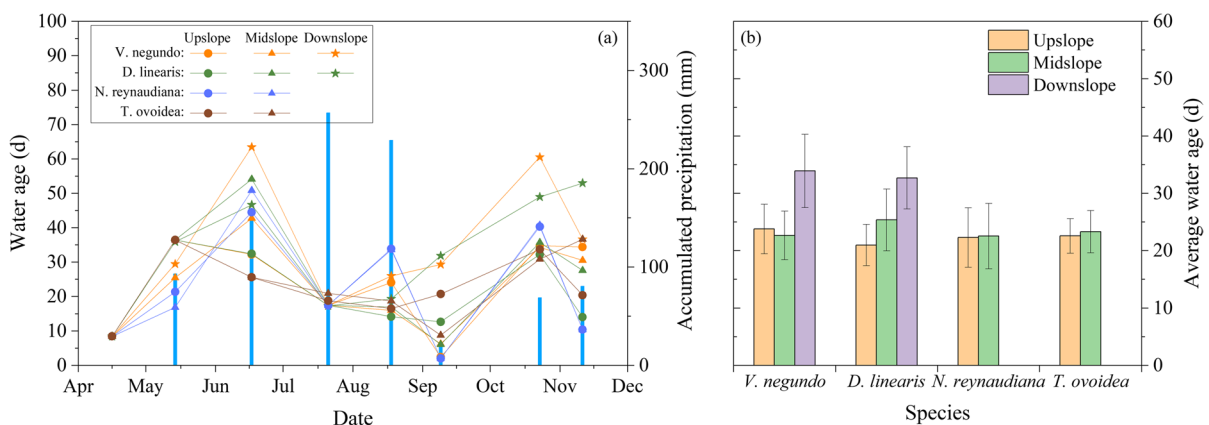


Fig. 5 Seasonal variations in water age of root water uptake for the four species on different slopes and interval precipitation (blue bars) between two consecutive sampling events (a).

The statistical average water age of root water uptake between species and between slopes is shown in (b)

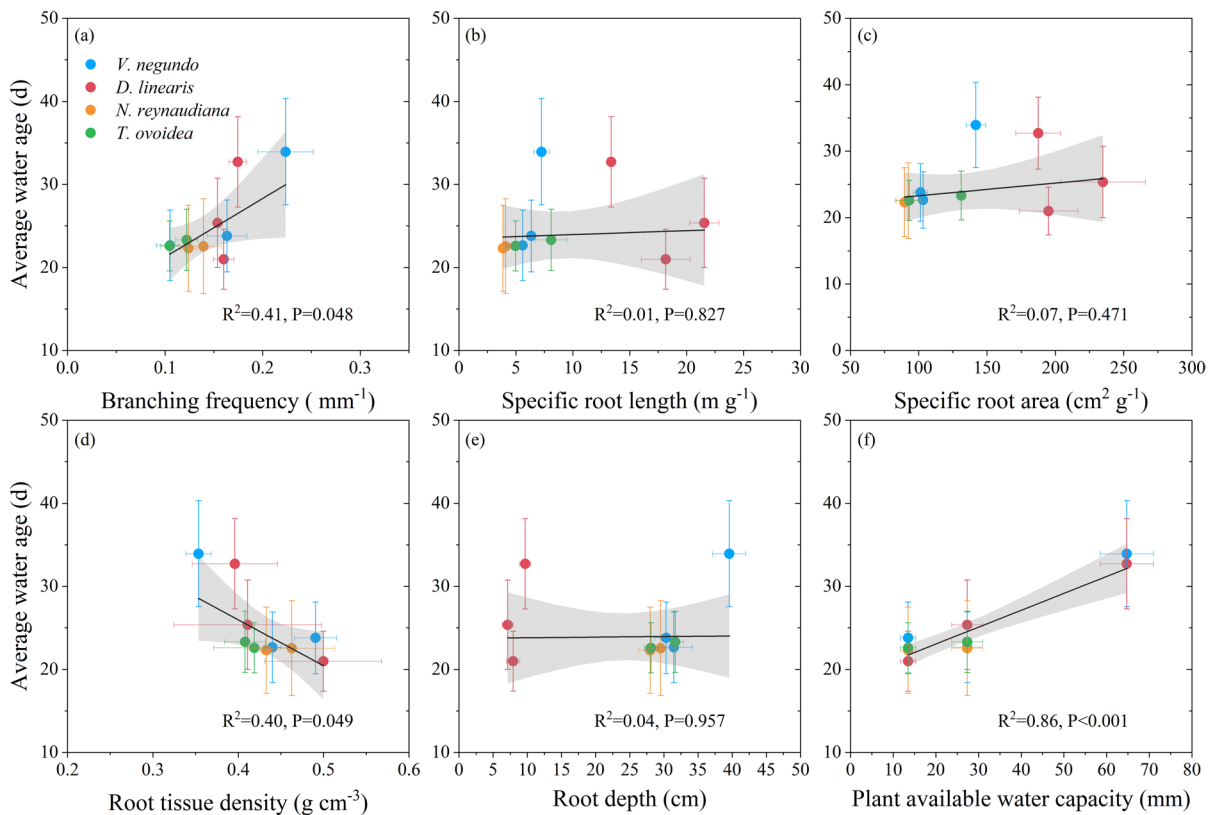


Fig. 6 Relationship between water age and the five root traits (a–e) and plant available water capacity (f)

positions, which has also been reported previously in karst ecosystem (Du et al. 2019; Ma et al. 2020; Ni et al. 2015). Such shallow rooting profiles are closely related to the subsurface soil-rock structures (Fig. 2). Soil thickness, the degree of bedrock weathering, and the combined structure of soil and bedrock are the key factors in forming the differences in soil-rock structures, which in turn exert a significant influence on the distribution of root systems. The heterogeneity of soil-rock structures can provide diverse ecological niches, serving as a foundation for plant root establishment and foraging. In karst regions, apart from soil, the soil-rock interfaces and bedrock fissures are found to be crucial pathways or reservoirs for water movement or storage (Zhang et al. 2023a, b). Consequently, roots will expand along these exploitable spaces.

Here, we summarize the impact of three distinct soil-rock structures, which are widely distributed in natural ecosystems, on root system architecture (Fig. 7). A critical aspect of such an impact is seen

in the influence on rooting depth. For situations with deep and continuous soil layers (Type 1), plant roots can spread freely and are primarily influenced by species-specific responses to soil compactness, with minimal impact from the bedrock. For the Type 2, the soil layer is thin, and the underlying bedrock has fewer developed fractures (weakly-weathered bedrock), presenting substantial resistance to root penetration. Consequently, roots are predominantly concentrated in the shallow layers (Fig. 1a, b), with well-developed lateral roots. Only a small fraction of roots can access these minor fissures (e.g., Fig. 4), as the water stored within them constitutes a critical potential water source for plants (Schwinning 2020). The results of this study belong to the second scenario, especially for species such as *V. negundo* and *T. ovoidea*, which expand their water and nutrient absorption area through lateral root extension. This predominantly lateral root distribution characteristic is more common in karst regions (Ma et al. 2020), particularly in areas dominated by dolomite. Of course, similar

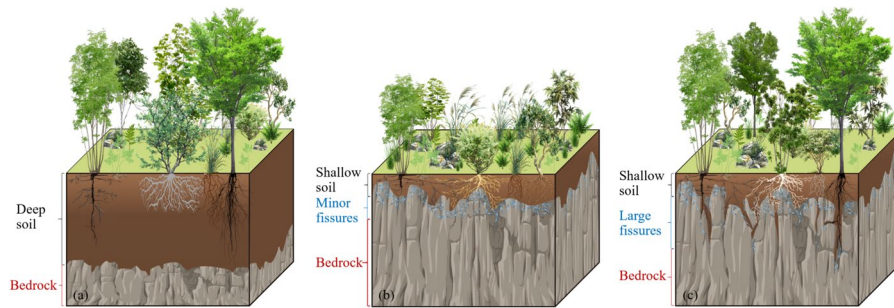


Fig. 7 Schematic of the effect of three typical soil-rock structures on root system architecture. (a) Type 1: deep and continuous soil layer, with deep bedrock depth, root system is generally spread freely and influenced by species-specific responses. (b) Type 2: shallow soils overlie weakly-weathered bedrock,

with only minor fissures in the upper part that store moisture (blue) but restrict root penetration. (c) Type 3: shallow soils overlie highly-weathered bedrock, with large fissures that allow soil, roots and water to move to deeper layers

reports have also been found in non-karst areas with shallow soil layers (Klimešová et al. 2018). However, when the underlying bedrock is highly weathered and has more developed large fissures (Type 3), plant roots can grow deeper along these fissures. This is often the case in limestone areas (Estrada-Medina et al. 2013; Leite et al. 2023). Especially in the soil-filled fissures, they serve as important channels for root extension, supporting larger woody plants (e.g., trees). This may be the key reason why, in the long-term process of vegetation succession, karst dolomite regions are primarily dominated by shrub and grass vegetation, whereas limestone regions are mainly covered by trees and shrubs (Jiang et al. 2020).

In addition to influencing root depth, soil-rock structures also profoundly affect the morphology of roots (curvature and branching). In habitats dominated by deep soils (e.g., Type 1 in Fig. 7a), the morphology of plant roots largely depends on genetic traits, as genetic differences often lead to distinct aggregation and specific branching patterns among different plants (Liu et al. 2019a, b; Salmela et al. 2020). For karst regions with soil-rock structures of the Type 2 and 3, or other similar soil-rock structures in non-karst regions, roots may direct their growth along shallow soil layers or through cracks. Particularly when there are many large gravel blocks in the soil, it significantly increases the curvature and branching frequency of roots. In this study, during the excavation of roots, it was found that the roots of *T. ovoidea* and *V. negundo* exhibited greater curvature and branching orders (Fig. 2) because they are adept at interweaving in the soil-rock interfaces or fractures

(Fig. 4). This demonstrates that these two plants possess strong resource acquisition capabilities. In contrast with our results, some studies have found that high gravel content can reduce root branching and specific root length (Hu et al. 2022; Semchenko et al. 2008). This may be related to gravel size or different responses between tree species. Although numerous studies have attempted to investigate effects of soil-rock structures on root system architecture, including effects on fine root biomass and allocation (Du et al. 2019; Jiang et al. 2018), the adaptive responses of root structure and morphology to the soil-rock-structure-induced heterogeneity and pressure remain an area that requires further in-depth research (Zhang et al. 2023a, b) due to the complex diversity of soil-rock structures and the plasticity of plant responses.

In karst areas, the soil-rock structures are usually extremely complex and hidden. In this study, a geophysical method (ERT) was used to detect the subsurface structures, which provided a good visualization of the spatial distribution of soil, bedrock, fissures, etc. and their combined structures (Luo et al. 2024; Tao et al. 2022). This helps to identify the possible distribution areas of roots in the subsurface and how the soil-rock structure affects the distribution of plant root system. Numerous studies have employed ground-penetrating radar (GPR) to explore the distribution of roots with different diameters due to its relatively high resolution of GPR for near-surface detection (Alani and Lantini 2020; Rodríguez-Robles et al. 2017; Zhang et al. 2024). Moreover, a combined application of GPR and ERT are able to detect distinctive microsites in shallow layer of

weathered bedrock suitable for tree root establishment (Rodríguez-Robles et al. 2017). This indicates that geophysical techniques (e.g., ERT, GPR) enable non-invasive 3D monitoring of root distribution and root-soil interactions, offering novel insights into plant root ecology. However, the resolution of fine-scale root architecture under conditions of complex soil-rock structures remains a challenge. In the future, more interdisciplinary research is still needed to advance our understanding in root ecology and soil-rock-root interactions.

Water use strategies and coexistence mechanisms of plants in soil-limited environment

Root systems adapt to variations in soil-rock moisture by dynamically adjusting the spatial distribution and temporal allocation of water acquisition sources. Due to the vertical and horizontal heterogeneity of root distribution, we did not explore the vertical root water uptake patterns, which is a shortcoming of this study. However, the observed maximum rooting depth in this study consistently demonstrates that these plants rely primarily on shallow water sources (0–40 cm). Maximum rooting depth directly regulates the potential and stability of root water uptake by determining the spatial extent of water available to plants (Nippert and Holdo 2015; Stocker et al. 2023). Although a recent study of global plant water uptake patterns suggests that maximum rooting depth is decoupled from water uptake depth (Bachofen et al. 2024), plant water use is mainly dependent on fine roots (Germon et al. 2020; Weemstra et al. 2016). During root excavation, the fine roots of the four species were found to be distributed in the soil, rock-soil interfaces, and small fissures (Fig. 4), suggesting that both soil water and rock/fissure moisture are important sources of water for the species at the study site. In addition, extensive lateral root spread was observed in these plants, which may reflect a common water use strategy for plants in soil-limited environments (Ma et al. 2020). A more spatial extensive lateral root system facilitates access to a larger potential pool of resources available to plants (Agee et al. 2021; Klimešová et al. 2018). Previous studies have reported that some species exhibit lateral root extension ranges exceeding tenfold the canopy radius (Meinen et al. 2009; Rewald and Leuschner 2009). This suggests a fundamental trade-off in root investment strategies within the physical

constraints of the shallow karst (dolomite-dominated) habitat. In shallow soils with underdeveloped cracks where deep root penetration is restricted, investing in horizontal root foraging is indeed more advantageous.

Differences in vertical root distribution and/or water uptake, commonly known as ecohydrological niche partitioning, have been demonstrated as a fundamental mechanism underlying plant coexistence (Kulmatiski et al. 2020; Weides et al. 2024; Zhao et al. 2024). For the four species investigated in this study, the entire root system of *D. linearis* is located within the top 10 cm soil layer, and differs obviously from the other three species in terms of vertical distribution of the root systems. This suggests that *D. linearis* probably does not compete with the other three species for soil or rock moisture below 10 cm. Furthermore, based on field observations and empirical investigations, we believe that rock/fissure-induced segregation of ecohydrological niches constitutes another critical mechanism driving plant coexistence in environments with shallow soils and complex soil-rock structures. This is because plant roots seek this source of water by penetrating into different bedrock fissures, and the isolating effect of bedrock may reduce interspecific water competition. The differential utilization of water from bedrock fissures and soil-rock interfaces, however, represents a critical niche partitioning mechanism. Such water source may buffer plants (e.g., shrubs in this study) against the rapid drying of the shallow soil, which can effectively stabilize species coexistence in this water-limited karst environment. There is increasing evidence that rock moisture utilization is an important survival strategy for plants (Bert et al. 2025; McCormick et al. 2021; Schwinning 2020), particularly given that access to deep water sources or bedrock-derived water resources during drought periods can significantly mitigate drought stress impacts (Ding et al. 2021; Hahm et al. 2020). However, for most plants, while the majority of their root systems are predominantly clustered within shallow layers, the functional contributions of limited root biomass in deep or bedrock fissures, along with interspecific variations in such adaptations, remain understudied in ecological and ecohydrological contexts.

The temporal dimension of root water uptake sources, as distinct from their spatial origins,

provides critical insights into plant adaptive strategies in response to precipitation variability. Our investigation revealed that the studied species exhibit relatively short averaged water ages of root water uptake (22–37 days), suggesting that these species are mainly dependent on water sources replenished by recent precipitation. These findings align with the results reported previously in karst ecosystem (Luo et al. 2023; Nie et al. 2012) and globally documented patterns where approximately 70% of plant water uptake primarily rely on water sources originated from current-month precipitation (Miguez-Macho and Fan 2021). Given that the precipitation in the humid karst regions is relatively high and frequent, and frequent precipitation helps to effectively replenish root zone water (Luo et al. 2021; Zhang et al. 2025), such an efficient exploitation of recent precipitation water by plants can facilitates multispecies coexistence in soil-limited habitats. However, under projected climate change scenarios with reduced precipitation regimes (Chen et al. 2021), species employing such rainfall water-dependent strategies may face elevated drought vulnerability due to diminished buffering capacity of shallow soil reservoirs.

Conclusion

The investigations of soil-rock structures, root architecture, and the water age dynamics of root water uptake for plants in soil-limited environments presented in this study offer significant new insights into the linkages among subsurface structures, root spatial deployment and plant water acquisition strategies. Key findings demonstrate: (1) Limited bedrock fissuring and shallow soil profiles (Table 2) constrain vegetation to shallow-rooted growth forms (maximum rooting depth < 45 cm), yet (2) heterogeneous soil-rock interfaces—characterized by patchy soil accumulations and variably weathered dolomite fractures—drive divergent root penetration strategies between lateral soil explorers (*V. negundo* and *T. ovoides*) and vertical fracture exploiters (*N. reynaudiana*). In addition, it was found that the fine-root branching frequency and tissue density became the key root traits controlling water age dynamics, suggesting that plants with higher fine root branching enabling access to water pockets of

varying temporal origins. These findings demonstrate that subsurface soil-rock structures act as a hidden driver of ecohydrological niche differentiation, where localized weathering creates water storage reservoirs that sustain multispecies coexistence through spatiotemporal complementarity in root water foraging. Future climate scenarios that change precipitation regimes could disrupt this delicate balance, as these plants predominantly rely on water sources originated from current-month precipitation (averaged water age < 33.9 d). The sustainability of such plant water adaptation strategies could be compromised by prolonged dry spells, as shallow-rooted/recent water using species are likely to die first if precipitation events become less frequent.

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Author contributions ZL, YN and HC conceived the ideas and designed methodology; ZL performed experiments and collected the data; ZL and DZ measured root traits with the help of FW, JL and HD; ZL analyzed the data, executed the calculations, and wrote the manuscript with the help of JZ, WL and HC. All authors contributed to the writing and editing of the manuscripts.

Data availability Data will be made available on request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Agee E, He L, Bisht G, Couvreur V, Shahbaz P, Meunier F, Gough CM, Matheny AM, Bohrer G, Ivanov V (2021) Root lateral interactions drive water uptake patterns under water limitation. *Adv Water Resour* 151:103896
- Alani AM, Lantini L (2020) Recent advances in tree root mapping and assessment using non-destructive testing methods: a focus on ground penetrating radar. *Surv Geophys* 41:605–646
- Bachofen C, Tumber-Davila SJ, Mackay DS, McDowell NG, Carminati A, Klein T, Stocker BD, Mencuccini M,

- Grossiord C (2024) Tree water uptake patterns across the globe. *New Phytol* 242:1891–1910
- Belovitch M, Brantley S, Aubrey DP (2022) Interspecific variation in the timing and magnitude of hydraulic redistribution in a forest with distinct water sources. *Plant Soil* 472:451–464
- Benettin P, Tagliavini M, Andreotti C, di Villahermosa FSM, Verdone M, Dani A, Penna D (2025) Ecohydrological dynamics and temporal water origin in a European Mediterranean vineyard. *Ecohydrol* 18:e2711
- Bert SD, Tomasella M, Duddek P, Natale S, Petruzzellis F, Carminati A, Zini L, D'Amico L, Tromba G, Nardini A (2025) Hydraulic properties of a rock-soil-root system: insights from *Fraxinus ornus* L. saplings growing on different carbonate rocks. *Plant Cell Environ* 10:1–11
- Brinkmann N, Seeger S, Weiler M, Buchmann N, Eugster W, Kahmen A (2018) Employing stable isotopes to determine the residence times of soil water and the temporal origin of water taken up by *Fagus sylvatica* and *Picea abies* in a temperate forest. *New Phytol* 219:1300–1313
- Brooks JR, Barnard HR, Colombe R, McDonnell JJ (2010) Ecohydrologic separation of water between trees and streams in a Mediterranean climate. *Nat Geosci* 3:100–104
- Chen L, Wang G, Miao L, Gnyawali KR, Li S, Amankwah SOY, Huang J, Lu J, Zhan M (2021) Future drought in CMIP6 projections and the socioeconomic impacts in China. *Int J Climatol* 41:4151–4170
- Chong G, Hai Y, Zheng H, Xu W, Ouyang Z (2021) Characteristics of changes in karst rocky desertification in southern and western China and driving mechanisms. *Chin Geogr Sci* 31:1082–1096
- Dai J, Zhang X, Wang L, Wang R, Luo Z, He X, Rao Z (2023) Water stable isotope characteristics and water use strategies of co-occurring plants in ecological and economic forests in subtropical monsoon regions. *J Hydrol* 621:129565
- Dawson TE (1996) Determining water use by trees and forests from isotopic, energy balance and transpiration analyses: the roles of tree size and hydraulic lift. *Tree Physiol* 16:263–272
- Ding Y, Nie Y, Chen H, Wang K, Querejeta JJ (2021) Water uptake depth is coordinated with leaf water potential, water-use efficiency and drought vulnerability in karst vegetation. *New Phytol* 229:1339–1353
- Du H, Liu L, Su L, Zeng F, Wang K, Peng W, Zhang H, Song T (2019) Seasonal changes and vertical distribution of fine root biomass during vegetation restoration in a Karst area, Southwest China. *Front Plant Sci* 9:2001
- Estrada-Medina H, Graham RC, Allen MF, Jiménez-Osorio JJ, Robles-Casolco S (2013) The importance of limestone bedrock and dissolution karst features on tree root distribution in northern Yucatán, México. *Plant Soil* 362:37–50
- Evaristo J, Jasechko S, McDonnell JJ (2015) Global separation of plant transpiration from groundwater and streamflow. *Nature* 525:91–94
- Evaristo J, Kim M, Haren J, Pangle LA, Harman CJ, Troch PA, McDonnell JJ (2019) Characterizing the fluxes and age distribution of soil water, plant water, and deep percolation in a model tropical ecosystem. *Water Resour Res* 55:3307–3327
- Fu X, Meng S, Kou L, Dai X, Wang H (2022) Multiple responses of fine root resource uptake strategies to gravel content in a subtropical plantation. *Can J Forest Res* 52:293–300
- Germon A, Laclau J, Robin A, Jourdan C (2020) Tamm review: deep fine roots in forest ecosystems: why dig deeper? *For Ecol Manage* 466:118135
- Gessler A, Bächli L, Rouholahnejad Freund E, Treydt K, Schaub M, Haeni M, Weiler M, Seeger S, Marshall J, Hug C, Zweifel R, Hagedorn F, Rigling A, Saurer M, Meusburger K, Sveriges L (2022) Drought reduces water uptake in beech from the drying topsoil, but no compensatory uptake occurs from deeper soil layers. *New Phytol* 233:194–206
- Hahm WJ, Rempe DM, Dralle DN, Dawson TE, Dietrich WE (2020) Oak transpiration drawn from the weathered bedrock vadose zone in the summer dry season. *Water Resour Res* 56:e2020WR027419
- Hu H, Bao W, Eissenstat DM, Huang L, Liu J, Li F (2022) Rock fragment content in soils shift root foraging behavior in xerophytic species. *Plant Soil* 478:671–688
- Hu H, Bao W, Huang L, Li F (2024) Shifting patterns in fine root distribution of four xerophytic species across soil structural gradients and years of growth. *Ecol Evol* 14:e10889
- Jia G, Liu Z, Chen L, Yu X (2017) Distinguish water utilization strategies of trees growing on earth-rocky mountainous area with transpiration and water isotopes. *Ecol Evol* 7:10640–10651
- Jiang P, Wang H, Fu X, Dai X, Kou L, Wang J (2018) Elaborate differences between trees and understory plants in the deployment of fine roots. *Plant Soil* 431:433–447
- Jiang Z, Liu H, Wang H, Peng J, Meersmans J, Green SM, Quine TA, Wu X, Song Z (2020) Bedrock geochemistry influences vegetation growth by regulating the regolith water holding capacity. *Nat Commun* 11:2392
- Klimešová J, Martínková J, Herben T (2018) Horizontal growth: an overlooked dimension in plant trait space. *Perspect Plant Ecol Evol Syst* 32:18–21
- Korboulewsky N, Tétégan M, Samouelian A, Cousin I (2020) Plants use water in the pores of rock fragments during drought. *Plant Soil* 454:35–47
- Kulmatiski A, Adler PB, Foley KM (2020) Hydrologic niches explain species coexistence and abundance in a shrub-steppe system. *J Ecol* 108:998–1008
- Leite PAM, Schmidt LM, Rempe DM, Olariu HG, Walker JW, McInnes KJ, Wilcox BP (2023) Woody plant encroachment modifies carbonate bedrock: field evidence for enhanced weathering and permeability. *Sci Rep*. <https://doi.org/10.1038/s41598-023-42226-7>
- Liu C, Xiang W, Zou L, Lei P, Zeng Y, Ouyang S, Deng X, Fang X, Liu Z, Peng C (2019a) Variation in the functional traits of fine roots is linked to phylogenetics in the common tree species of Chinese subtropical forests. *Plant Soil* 436:347–364
- Liu H, Jiang Z, Dai J, Wu X, Peng J, Wang H, Meersmans J, Green SM, Quine TA (2019b) Rock crevices determine woody and herbaceous plant cover in the karst critical zone. *Sci China Earth Sci* 62:1756–1763
- Liu W, Behzad HM, Luo Z, Huang L, Nie Y, Chen H (2025) Species-specific root distribution and leaf iso/

- anisohydric tendencies shape transpiration patterns across heterogeneous karst habitats. *Plant Cell Environ* 48:199–212
- Liu W, Nie Y, Luo Z, Wang Z, Huang L, He F, Chen H (2023) Transpiration rates decline under limited moisture supply along hillslopes in a humid karst terrain. *Sci Total Environ* 894:164977
- Luo Z, Guan H, Zhang X, Xu X, Dai J, Hua M (2019) Examination of the ecohydrological separation hypothesis in a humid subtropical area: comparison of three methods. *J Hydrol* 571:642–650
- Luo Z, Lian J, Nie Y, Zhang W, Wang F, Huang L, Chen H (2024) Improving soil thickness estimations and its spatial pattern on hillslopes in karst forests along latitudinal gradients. *Geoderma* 441:116749
- Luo Z, Nie Y, Chen H, Guan H, Zhang X, Wang K (2023) Water age dynamics in plant transpiration: the effects of climate patterns and rooting depth. *Water Resour Res* 59:e2022WR033566
- Luo Z, Nie Y, Ding Y, Chen H (2021) Replenishment and mean residence time of root-zone water for woody plants growing on rocky outcrops in a subtropical karst critical zone. *J Hydrol* 603:127136
- Ma X, Chen H, Nie Y (2020) Common species maintain a large root radial extent and a stable resource use status in soil-limited environments: a case study in subtropical China. *Front Plant Sci* 11:1260
- McCormick EL, Dralle DN, Hahm WJ, Tune AK, Schmidt LM, Chadwick KD, Rempe DM (2021) Widespread woody plant use of water stored in bedrock. *Nature* 597:225–229
- Meinen C, Leuschner C, Ryan NT, Hertel D (2009) No evidence of spatial root system segregation and elevated fine root biomass in multi-species temperate broad-leaved forests. *Trees* 23:941–950
- Miguez-Macho G, Fan Y (2021) Spatiotemporal origin of soil water taken up by vegetation. *Nature* 598:624–628
- Montaldo N, Corona R, Currelli M, Sirigu S, Piroddi L, Oren R (2021) Rock water as a key resource for patchy ecosystems on shallow soils: digging deep tree clumps subsidize surrounding surficial grass. *Earths Future* 9:e1870E-e2020E
- Nardini A, Petruzzellis F, Marusig D, Tomasella M, Natale S, Altobelli A, Calligaris C, Floriddia G, Cucchi F, Forte E, Zini L (2021) Water “on the rocks”: a summer drink for thirsty trees? *New Phytol* 229:199–212
- Nardini A, Tomasella M, Di Bert S (2024) Bedrock: the hidden water reservoir for trees challenged by drought. *Trees* 38:1–11
- Ni J, Luo DH, Xia J, Zhang ZH, Hu G (2015) Vegetation in karst terrain of southwestern China allocates more biomass to roots. *Solid Earth* 6:799–810
- Nie Y, Chen H, Wang K, Yang J (2012) Water source utilization by woody plants growing on dolomite outcrops and nearby soils during dry seasons in karst region of southwest China. *J Hydrol* 420–421:264–274
- Nippert JB, Holdo RM (2015) Challenging the maximum rooting depth paradigm in grasslands and savannas. *Funct Ecol* 29:739–745
- Rewald B, Leuschner C (2009) Belowground competition in a broad-leaved temperate mixed forest: pattern analysis and experiments in a four-species stand. *Eur J Forest Res* 128:387–398
- Rodríguez-Robles U, Arredondo T, Huber-Sannwald E, Ramos-Leal JA, Yépez EA (2017) Technical note: application of geophysical tools for tree root studies in forest ecosystems in complex soils. *Biogeosciences* 14:5343–5357
- Salmela MJ, Velmala SM, Pennanen T (2020) Seedling traits from root to shoot exhibit genetic diversity and distinct responses to environmental heterogeneity within a tree population. *Oikos* 129:544–558
- Schwinning S (2013) Do we need new rhizosphere models for rock-dominated landscapes? *Plant Soil* 362:25–31
- Schwinning S (2020) A critical question for the critical zone: how do plants use rock water? *Plant Soil* 454:49–56
- Semchenko M, Zobel K, Heinemeyer A, Hutchings MJ (2008) Foraging for space and avoidance of physical obstructions by plant roots: a comparative study of grasses from contrasting habitats. *New Phytol* 179:1162–1170
- Sprenger M, Leistert H, Gimbel K, Weiler M (2016) Illuminating hydrological processes at the soil-vegetation-atmosphere interface with water stable isotopes. *Rev Geophys* 54:674–704
- Sprenger M, Stumpp C, Weiler M, Aeschbach W, Allen ST, Benettin P, Dubbert M, Hartmann A, Hrachowitz M, Kirchner JW, McDonnell JJ, Orlowski N, Penna D, Pfahl S, Rinderer M, Rodriguez N, Schmidt M, Werner C (2019) The demographics of water: a review of water ages in the critical zone. *Rev Geophys* 57:800–834
- Stocker BD, Tumber-Dávila SJ, Konings AG, Anderson MC, Hain C, Jackson RB (2023) Global patterns of water storage in the rooting zones of vegetation. *Nat Geosci* 16:250–256
- Tao M, Chen X, Cheng Q, Binley A (2022) Evaluating the joint use of GPR and ERT on mapping shallow subsurface features of karst critical zone in southwest China. *Vadose Zone J* 21:e20172
- Tong X, Brandt M, Yue Y, Horion S, Wang K, Keersmaecker WD, Tian F, Schurgers G, Xiao X, Luo Y, Chen C, Myneni R, Shi Z, Chen H, Fensholt R (2018) Increased vegetation growth and carbon stock in China karst via ecological engineering. *Nat Sustain* 1:44–50
- Wang B, Yan C, Shi Z, Ding J, Zhang T, Qin L, Qiu GY (2023a) Seasonal variation in water uptake patterns of two greening species and their responses to rainfall events in a subtropical megacity of China. *J Hydrol* 618:129262
- Wang J, Fu B, Lu N, Zhang L (2017) Seasonal variation in water uptake patterns of three plant species based on stable isotopes in the semi-arid Loess Plateau. *Sci Total Environ* 609:27–37
- Wang J, Wang C, Li H, Liu Y, Li H, Ren R, Si B (2023b) Rock water use by apple trees affected by physical properties of the underlying weathered rock. *Agric Water Manage* 287:108413
- Wang K, Zhang C, Chen H, Yue Y, Zhang W, Zhang M, Qi X, Fu Z (2019) Karst landscapes of China: patterns, ecosystem processes and services. *Landsc Ecol* 34:2743–2763
- Weemstra M, Mommer L, Visser EJW, van Ruijven J, Kuyper TW, Mohren GMJ, Sterck FJ (2016) Towards a multidimensional root trait framework: a tree root review. *New Phytol* 211:1159–1169

- Weides SE, Hájek T, Liancourt P, Herberich MM, Kramp RE, Tomiolo S, Pacheco Riaño LC, Tielbörger K, Májeková M (2024) Belowground niche partitioning is maintained under extreme drought. *Ecology* 105:e4198
- Wu H, Zhao G, Li X, Wang Y, He B, Jiang Z, Zhang S, Sun W (2019) Identifying water sources used by alpine riparian plants in a restoration zone on the Qinghai-Tibet Plateau: evidence from stable isotopes. *Sci Total Environ* 697:134092
- Yang B, Wen X, Sun X (2015) Seasonal variations in depth of water uptake for a subtropical coniferous plantation subjected to drought in an East Asian monsoon region. *Agr Forest Meteorol* 201:218–228
- Yin D, Gou X, Liu J, Zhang D, Wang K, Yang H (2024) Increasing deep soil water uptake during drought does not indicate higher drought resistance. *J Hydrol* 630:130694
- Yue Y, Wang L, Zhang X, Wang K (2024) Towards achieving carbon neutrality: the role of vegetation restoration in karst regions of southwest China. *J Earth Sci-China* 35:1044–1048
- Zhang J, Chen H, Fu Z, Wang F, Wang K (2023a) Towards hydrological connectivity in the karst hillslope critical zone: insight from using water isotope signals. *J Hydrol* 617:128926
- Zhang J, Wang S, Fu Z, Wang F, Wang K, Chen H (2025) Regulation of preferential flow by soil thickness on small hillslopes with complex topography through intensive high-frequency soil moisture monitoring. *Geophys Res Lett* 52:e2024GL112674
- Zhang L, Zhang Z, Guo L, Cui X, Butnor JR, Li S, Cao X, Chen X (2024) A simple method for estimating the coarse lateral root biomass of shrubs using ground-penetrating radar: validation by *Caragana microphylla* Lam. in Inner Mongolia. *Sci Total Environ* 919:170897
- Zhang Z, Fan B, Song C, Zhang X, Zhao Q, Ye B (2023b) Advances in root system architecture: functionality, plasticity, and research methods. *J Resour Ecol* 14:15
- Zhao Y, Wang L, Chun KP, Ziegler AD, Evaristo J (2024) Dynamic hydrological niche segregation: how plants compete for water in a semi-arid ecosystem. *J Hydrol* 630:130677

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